Targetted Requirements for Macroeconomic Stabilization

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Introduction

Reserve requirements (RR) as tool for macroeconomic stabilization

Focus on China's experiences: PBOC has used targeted RR adjustments to support smaller banks

Motivation:
- Small vs. Large banks regulations: **Additional MaP requirements on Large Banks** $\rightarrow$ e.g., SIFIs regulations
- China: Larger RR cuts for small and local banks relative to National Banks in past crises

Question: How to Rationalize this policy design?

Empirical findings: Targeted RR changes affect loan growth rates and overall economic output

Framework: Model where firms can borrow from local (L) or national (N) banks affected by type-specific RR:

- For local banks: $\downarrow RR_L \rightarrow \uparrow$ (leverage, intermediation and output)
- For national banks: ? (conflicting effects at play — expansionary policy may have recessionary effects)

Trade-off: $\downarrow$ RR can stabilize economy but increases leverage and default risk ($\downarrow$ financial instability)

Effect of RR is considerable but also has high potential for unintended financial instability effects

Overall thoughts

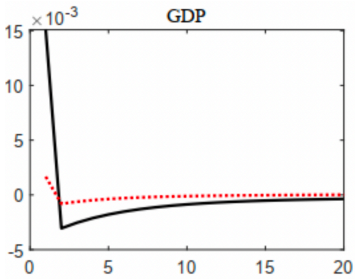
I liked the paper! It relates closely with the financial frictions literature while covering a new question

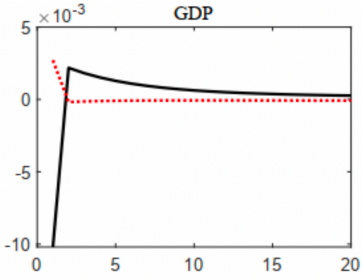
The main insight:

The same prudential policy may have completely opposite stabilization effects depending on its target

⇒ It is key to choose the target wisely

Effects of lower RR in local banks (left) v.s national banks (right):





Model

- Two types of banks: National and Local
  - National: Better liquidity provision (due to size)
  - Local: Better local monitoring of lenders → Handles better friction costs ↓ EFP
- Bernanke, Gilchrist, Gertler (1999) type of framework: Frictions take place at Firms
- Firms face costs from switching banks $\Rightarrow$ they migrate lenders only upon large shocks
- Government sets RR at the bank type level
  - Government also provides deposit insurance for all savers

Model (cont.)

Some selected features of the model:

Households:

National Deposits in Utility preferences.

$$U = \mathbb{E}_t \sum_{s=0}^{\infty} \left[\ln(C_{t+s}) - \Psi_h \frac{H_{t+s}}{1+\eta} + \Psi_n \ln(D_{n,t+s}) \right]$$

Assets' Euler Equations:

Asymmetric asset choices consequences

$$1 = \mathbb{E}_t \beta R^d_{nt} \frac{\Lambda_{t+1}}{\Lambda_t} + \Psi_n \frac{1}{\Lambda_t D_{n,t}},$$

$$1 = \mathbb{E}_t \beta R^d_{lt} \frac{\Lambda_{t+1}}{\Lambda_t}$$

This captures that: National Banks provide liquidity services, thus D_n enters utility as money would (MIU)

On average this implies that R_l is higher (see Euler Equations)

Model (cont.)

Banks:

Two types of banks n,l (national, local)

Banks compete for deposits and set a rate for loans s.t. CES type demand for deposits

Policy tool is a RR given by a share of deposits:

$$b_{b,t}(i) = (1 - \tau_{b,t})d_{b,t}(i)$$

Policy tool: RR (τ_b)

Profits are affected by RR (a part won't yield returns)

Deposits and rates chosen to maximize expected profits

$$\pi_t(i) = \int_{\bar{\epsilon}_{b,t}}^{\infty} [\tau_{b,t} d_{b,t}(i) + \epsilon_{b,t} R_{b,t} B_{b,t}(i) - r^d_{b,t}(i) d_{b,t}(i)] d\Phi(\epsilon_{b,t})$$

Model (cont.): Policy

Optimal lending decisions imply:

$$\frac{\theta_d}{\theta_d+1}\left[\frac{\int_{\bar{\epsilon}_{b,t}}^{\infty}\epsilon_{b,t}d\Phi(\epsilon_{b,t})}{1-\Phi(\bar{\epsilon}_{b,t}(i))}R_{b,t}(1-\tau_{b,t})+\tau_{b,t}\right]=r^d_{b,t}(i)$$

RHS: Marginal return on lending; LHS: Marginal cost

$\Rightarrow$ **higher RR lowers banking marginal returns**

Analogous to other MaP tools in the literature: Banerjee, Devereux, Lombardo (2017), Agenor et al (2022), Benigno, Aoki, Kiyotaki (2018), among others.

Optimal Policy Design Exercise:

- Policy setup as RR rules (Taylor type)
- Both RR's are allowed to change depending on business cycle

$$\tau_{l,t}=\bar{\tau}_l+\psi_{ly}\ln\left(G\tilde{D}P_t\right)$$

$$\tau_{n,t}=\bar{\tau}_n+\psi_{ny}\ln\left(G\tilde{D}P_t\right)$$

Specific policy parameter for each bank type

Mechanism

Usual models: any MaPP (tightening) $\rightarrow$ cools down aggregate fin. intermediation.

This paper:

- **Lower N banks RR**

Deteriorates Fin. Stability and Cycle!
(worse of both worlds)

- Effect on Leverage (intensive mg): N banks intermediate more $\Rightarrow$ higher leverage
- Effect on EFP (extensive mg): Firms migrate to N Banks and mean EFP increases $\Rightarrow$ lower leverage

Net effect: (may depend on parameters) lower leverage from national banks RR cut $\Rightarrow$ **Recessionary!**

- **Lower L banks RR**

- Effect on Leverage (intensive mg): L banks intermediate more $\Rightarrow$ higher leverage
- Effect on EFP (extensive mg): Firms migrate to L banks $\Rightarrow$ lower average EFP (better monitoring)

$\Rightarrow$ Increase in avg. leverage too $\Rightarrow$ Expansionary Intended Effects (at Fin. Stab cost)

Key lesson: A badly designed policy may lead to unintended outcomes

Comments: Big picture remarks

- The Focus of the paper is China and thus conclusions may not generalize.

- **Long-term effects:** Paper focuses on the short-term policy trade-off only (stabilization vs. fin. stability)

Can be complemented with an Long-run assessment (welfare)

Should MaPP be used for stabilization too or only for financial stability purposes? (or to what extent)
[More on how to explore this in current setup —w/o welfare accounting— on specific comments below]

- **Other policymakers:** Paper focuses on financial regulators whose primary concern is not stabilization

how beneficial is to carry out stabilization with financial regulation rather than with other policy tools ?

how relevant if other policymakers already carry out stabilization ? (perhaps critical at ZLB)

The main result implies a different financial stability vs. stabilization trade-off for each bank type

However, these trade-offs may change with cyclical and monetary conditions (that also affect EFP)
(e.g., Rey, Coimbra, 2023 ReStud)

Other comments: Specific to current setup

- There is no exploration of **optimal policy-making**:

To what extent pursue stabilization and when switch attention to financial stability issues?

Reduced form possible approach (low cost): Mixed RR rule (as opposed to current single-argument rule)

$$\tau_{l,t} = \bar{\tau}_l + \psi_{ly} \ln\left(G\tilde{D}P_t\right) + \phi_{l,fs}\text{FinStabIndicator}_t$$

$$\tau_{n,t} = \bar{\tau}_n + \psi_{ny} \ln\left(G\tilde{D}P_t\right) + \phi_{n,fs}\text{FinStabIndicator}_t$$

Possible choices: EFP, Credit/GDP, Leverage, AverageMonitoringCosts

Other comments (cont.)

- How would results differ if lenders are the ones depicting financial (agency) frictions?
(setup: Gertler and Karadi, 2011, and related literature)

- Effects of MaP are stronger from instruments targeting borrowers.
- Core of crises: Financial Intermediation Sector (instead of firms' lending constraints)
- Easier to motivate: Different degrees of agency costs in L vs. N banks.
- Can even consider interbank market: L banks getting funds from N banks too

More intuitive if tool targets source of frictions

- More than one macroprudential tool under study: Deposit insurance.

Interesting to determine to what extent it affects the stabilization properties of regulations.

Does it hinder, replace, or complement the role or RR? [Does it elicit frictions in banks?]

Other comments (cont.)

- Asymmetry: Is trade-off between stabilization and financial stability similar along business cycle as argued? (e.g., in recessions vs. booms? and during ZLB and out of it)

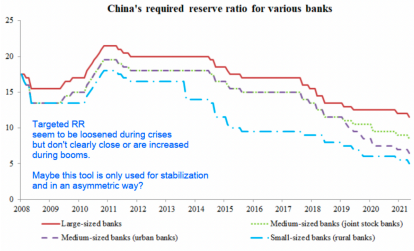
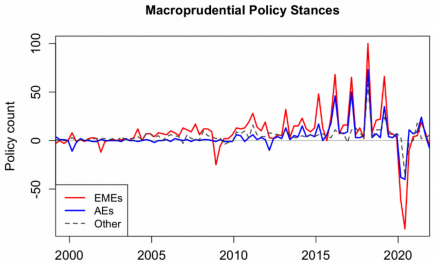


FIGURE 1. China's required reserve ratios for banks of different sizes.



Left: RR in China; Right: All Global Macroprudential Interventions (source: iMaPP-IMF)

MaP Tightenings have stronger effects than loosenings (Richter, Schularick and Shim, 2019)

Rest of toolkit comoves strongly with the cycle while RR gap only seems to widen

Worth considering in the model or exploring further empirically?

Other comments (cont.)

Banking setup with Monopsonistic competition could be clarified further:

It's not clear who chooses optimally the deposit rate $r^d_{b,t}(i)$:

The paper states "The banks choose deposits and loans to maximize expected profits subject to flow of funds constraint (policy)." However, given the deposits, the RR (flow-of-funds constraint) dictates the loans, so it's not too clear how or why these are separate decisions and where the rate is determined.

Also, not clear why we should consider this market structure.

Does the banking market work like a monopsony in reality? If so, it would be great to highlight that!

Other papers have, instead, favored a monopolistic framework and document how it can augment financial frictions (Mandelman, 2010 JIE)

Conclusions

Really nice read, I learned a lot!

This is a good paper that tackles the task of adding an interesting layer to the discussion of macroprudential policy effects and financial frinctions

It could help answering to what extent and when Stabilization becomes a more relevant objective for RR adjustments compared to financial stability

There is some scope for further normative policy analysis and assumption clarifications/justifications

I look forward to reading future versions of this paper and other work by Mark, Zheng and Jingyi

Thank You!